

JUN YAMAMOTO

PHD CANDIDATE IN NETWORK SCIENCE

Department of Network and Data Science
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RESEARCH PROFILE

I study organizational principles of network structure and dynamics using statistical physics. My work focuses on physical networks, coarse-graining and scale-invariance, critical phenomena, diffusion, and synchronization, with the aim of developing a theory of dynamical processes in physical networks.

EDUCATION

- 2023–2027 (expected) **PhD Network Science**, Central European University, Vienna, Austria
Supervisors: Prof. Dr. Márton Pósfai and Prof. Dr. János Kertész
Thesis: “Dynamical processes on physical networks” (tentative).
- 2026/06–07 AccelNet-MultiNet Fellowship Research Visit, Center for Complex Networks
Research, Northeastern University, Boston, USA
Host PI: Prof. Dr. Albert-László Barabási
- 2022–2023 **MSc Mathematics**, Queen Mary University of London, London, UK
Supervisor: Prof. Dr. Ginestra Bianconi
Thesis: “Modelling higher-order network dynamics in the presence of triadic interactions.”
- 2017–2022 **BEng Applied Physics**, Hokkaido University, Sapporo, Japan
Supervisor: Prof. Dr. Kousuke Yakubo
Thesis: “Bifractality of fractal scale-free networks.”
- 2019–2020 Exchange Studies, Department of Physics, ETH Zürich, Zürich, Switzerland

PUBLICATIONS

1. M. Niedostatek, A. Baptista, **J. Yamamoto**, J. Kurths, R. Sanchez Garcia, B. MacArthur, and G. Bianconi, Mining higher-order triadic interactions. *Nature Communications* **16**, 11613 (2025). DOI: 10.1038/s41467-025-66577-z
2. K. Yakubo, G. Shimojo, and **J. Yamamoto**, Random walks on bifractal networks. *Physical Review E* **110**, 064318 (2024). DOI: 10.1103/PhysRevE.110.064318
3. **J. Yamamoto** and K. Yakubo, Bifractality of fractal scale-free networks. *Physical Review E* **108**, 024302 (2023). DOI: 10.1103/PhysRevE.108.024302

WORKING PAPERS

1. **J. Yamamoto**, I. Bonamassa, and M. Pósfai. Localization transitions of diffusion dynamics in physical networks.
2. S. Mizutaka, **J. Yamamoto**, and K. Yakubo. Structural crossover of complex networks: bridging degree correlation and fractality.

CONFERENCE PRESENTATIONS

- 2025 **J. Yamamoto**, I. Bonamassa, and M. Pósfai. *Localization-delocalization transition in physical networks*. The 29th International Conference on Statistical Physics (StatPhys29), Florence, Italy. July 14, 2025.
- J. Yamamoto**, I. Bonamassa, and M. Pósfai. *Delocalization of physical Laplacian eigenvectors due to degree-volume correlation*. The International School and Conference on Network Science (NetSci2025), Maastricht, the Netherlands. June 4, 2025.
- 2022 **J. Yamamoto** and K. Yakubo. *Bifractal property of stochastic scale-free networks*. The Physical Society of Japan 77th Annual Meeting, online. March 15, 2022.
- 2021 **J. Yamamoto** and K. Yakubo. *Multifractality of scale-free networks*. The Physical Society of Japan 76th Annual Meeting, online. September 20, 2021.

CONFERENCE POSTERS

- 2026 **J. Yamamoto**, I. Bonamassa, and M. Pósfai, *When node weight matters: spectral transitions in physical networks*. Network Science Seminar 2025, Kyoto, Japan, February 17–20, 2026.
- 2022 **J. Yamamoto**, G. Shimojo, and K. Yakubo, *Bifractal nature of fractal scale-free networks and its implications*. Network Science Seminar 2022, Kyoto, Japan, August 23–25, 2022.
- 2021 **J. Yamamoto** and K. Yakubo, *Bifractality of scale-free networks*. Network Science Seminar 2021, Kanazawa, Japan, December 11–12, 2021. *Poster Award*.

SEMINARS AND WORKSHOPS

- 2026 **J. Yamamoto**, I. Bonamassa, and M. Pósfai, *Eigenmode Localization in Physical Network Laplacians and Its Implications for Synchronization*. Group seminar at the Nakao Lab, Science Tokyo, Tokyo, Japan. February 9, 2026.
- 2025 **J. Yamamoto**, I. Bonamassa, and M. Pósfai, *Physical Networks as Network-of-Networks*. ネットワーク科学勉強会（運営：ネットワーク生態学研究会）, Online. May 25, 2025.
- J. Yamamoto**, I. Bonamassa, and M. Pósfai, *Reentrant localization in physical Laplacian eigenvectors*. DYNASNET workshop on graphs, networks, and brains, Budapest, Hungary. May 13, 2025.
- 2024 **J. Yamamoto**, I. Bonamassa, and M. Pósfai. *Laplacian Spectra and Angular Constraints in Physical Networks*. Annual DYNASNET Meeting, Valtice, Czech Republic. October 16, 2024.

FELLOWSHIPS AND GRANTS

- 2026 AccelNet-MultiNet Fellowship, National Science Foundation. Research visit to the Center for Complex Networks Research, Northeastern University, Boston, USA, to work with Prof. Dr. A.-L. Barabási. (\$4,000).
- 2022–2023 ITO Scholarship for Japanese Students, ITO Foundation. Tuition support, monthly stipend, and travel allowance for master’s study in the UK. (~\$50,000).

SELECTED AWARDS AND HONORS

- 2025 First-Year PhD Award, Central European University.
- 2023 Principal's Prize, Queen Mary University of London.
- 2022 Dean's Award for Academic Achievement, Hokkaido University.

TEACHING

- 2024–2026 Teaching Assistant, Central European University. Scientific Python (graduate); Introduction to Linear Algebra (undergraduate).
- 2021–2023 Teaching Assistant, Hokkaido University. Statistical Mechanics I; Applied Mathematics II; Computational Science; Introduction to Statistics.

ACADEMIC SERVICE

- 2025–present Ad hoc reviewer, *Nature Communications*; *Transactions of the Japanese Society for Artificial Intelligence*.
- 2025–2026 Organizer, Physical Networks IV Satellite, NetSci2026, Boston, USA.
- 2024–2025 Organizer, Physical Networks III Satellite, NetSci2025, Maastricht, the Netherlands.
- 2024–2025 Organizer, Data Stories 11th Edition, Department of Network and Data Science, Central European University.

OTHER EXPERIENCE

- 2022 Data Scientist, Fujitsu, DX Large Data Platform Business Group. Natural language processing and network analysis for large-scale Japanese text data.
- 2020 Summer Research Intern, Japan Atomic Energy Agency. Machine learning algorithms for accelerated molecular dynamics simulation of materials.
- 2018 Research Intern, Okinawa Institute of Science and Technology. Electron microscopy and vacuum system development for protein crystallography.

SKILLS

Programming Python, Julia, C++, Fortran.

Tools Linux, Git, L^AT_EX, Mathematica.

Languages Japanese (native), English (fluent), German (beginner).

REFEREES

Prof. Dr. Márton Pósfai, Department of Network and Data Science, Central European University, Vienna, Austria. *PhD supervisor*.

Prof. Dr. János Kertész, Department of Network and Data Science, Central European University, Vienna, Austria. *PhD co-supervisor*.

Prof. Dr. Ginestra Bianconi, School of Mathematical Sciences, Queen Mary University of London, London, UK. *MSc supervisor*.

Prof. Dr. Kousuke Yakubo, National Institute of Technology, Asahikawa College, Asahikawa, Japan. *Undergraduate supervisor*.